Gabryel Mason-Williams

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https://gmw99.github.io/

Research software associate who is passionate about machine learning and big data, with strong technical, research, business, and interpersonal skills for working in a team and completing novel research projects.

Education

- **Queen Mary University of London** **London**
  MSc Artificial Intelligence, Expected Distinction *2022-2023*
  Modules: Machine Learning, Artificial Intelligence, Natural Language Processing, Applied Statistics, Neural Networks and NLP, Neural Networks and Deep Learning, Information Retrieval, Ethics, Regulation and Law in Advanced Digital Information Processing and Decision Making, Dissertation Project
- **University of Plymouth** **Plymouth**
  BSc (Hons) Computer Science with a Year in Industry, 1st Class Honours (76.26%) *2017-2021*
  Final Year Modules: Computing Project (81.68%), Machine Learning (77.53%), Parallel Computing (75.60%), Alternative Paradigms (84%), Advanced Computing and Networking Infrastructures (78.05%)
  Deans List Award 2017-2018

Previous Employment

- **Rosalind Franklin Institute** **July 2022–Present**
  Research Software Associate *Contract remote*
  At the Rosalind Franklin Institute (RFI), I am presently involved in two research projects. DisTRaX, which I lead, is the productionisation and improvement of previous research and Kompressor, a machine learning project to create a novel neural losses compression algorithm for extensive volumetric scientific data. Presently in the Kompressor project, I have added the functionality to train over multiple nodes and GPUs in an HPC cluster. In this role, I also give talks at conferences [1], write papers and teach the PhD students.

  July 2021–July 2022
  Junior Research Software Engineer *Contract remote*
  In this role I conducted research in High-performance computing (HPC) for big data processing. Within this research, I lead the development and maintenance of two significant projects, DosNa [0] and DisTRaC [2] [3] . I was also involved in writing a successful grant bid for £10,000 in research funding for "Using Novel Cluster Technology to improve performance of Cryo EM analysis" with the University of Bristol. This was successful and lead to a reduction in processing times of 4.37%.
  As well I have given talks at UKRI Cloud Workshop 2022 [4] and Ceph Virtual 2022 based on the research I have done. In this role I also taught the PhD students about why machine learning techniques are used at RFI and how to effectively use HPC for research
- **Exeter Mathematics School** **October 2021–July 2022**
  Maths Circle Mentor *Contract remote*
  I mentor students in years 8 and 9 in mathematics to encourage the development of mathematical thinking outside of school context- the aim is to improve their ability to be adventurous, articulate and accepting by providing them challenging questions to become better mathematicians.
- **Diamond Light Source** **June 2019–September 2020**
  Scientific Computing Research Placement: Year in Industry Student *Oxford/Remote*
  At Diamond, I researched the use of "High Performance Object Stores for Intermediate Data processing", which involved using technologies such as High-Performance Compute Clusters, Univa Grid Engine, Savu, Ceph and relevant deployment tools as well as programming technologies such as MPI, C, python and bash. This research entailed: deploying and maintaining a semi-production Ceph cluster using ansible, making GRAM [5] a kernel module to represent RAM as a block device for Ceph and DisTRaC [3] an application that allows Ceph to run on to on a High Performance Computing cluster using RAM. I also presented a poster at Ceph Day CERN and gave a well-attended talk on DisTRaC to the Ceph Science Group.

Notable Projects

- **Final Year Dissertation 'Automating Lab Book Data Collection With Edge Machine Learning' - 2021**

  This is an interdisciplinary project in collaboration with the Rosalind Franklin Institute. The project's objective was to develop an automated way to collect lab book information and automatically record it to a central database using novel machine learning and computer vision techniques on edge computing, creating a searchable interactive digitised lab book. This project was developed in Python and was a success receiving a grade of 81.68%. The poster and dissertation is available: Poster, Dissertation

- **Visualisation of Objects Inside an Object Store in Realtime - 2019**

  I was a part of a team that created a Raspberry Pi Ceph compute cluster, which displayed tomography data processing in real-time for the SuperComputing 19 conference in Denver, STFC booth. For this project, I made the Python web application using flask. The web app allowed users to browse objects inside a pool and also view the slices of the tomography data processing as they are created and processed in real-time. A video of the process can be seen: here

- **A report and talk on 'Face recognition on social networking sites' - 2017**

  This was an individual task where I had to find an area of interest and see how far I could develop this within six months. The deliverable was a report and a 30-minute talk to a lay audience about the subject area. In the project, I created a piece of software using Python and OpenCV that could follow your head around if you were in view of the camera and discussed the potential of how you would go about collecting information about someone on social networking sites.

Technical and Personal skills

- **Programming Languages:** Python, C, Matlab, LaTeX, Bash.
- **Programming Frameworks:** Scientific Python Stack, Machine learning (Jax, Pytorch, Tensorflow), Distributed and Parallel Systems (Rados, Boto3, mpi4py, Ray) .
- **Industry Software Skills:** Research, Software Engineering, Machine Learning, Computer Vision, Object Orientated Programming, Object and File Storage, OpenMPI, High Performance Compute clusters, Linux, Version control, Unit testing, Data Visualisation, Agile Workflows.
- **General Business Skills:** Strong presentation, written, time management, problem-solving, and communication skills.
- **Other:** Effective at learning new frameworks, adaptability, disseminating knowledge via talks, posters and papers, research experience, experience working in and with interdisciplinary teams.

Publications and Talks

[1] G. Mason-Williams, "Distrac: Accelerating high performance data processing," English, Computing Insight UK 2022 : Sustainable HPC, CIUK ; Conference date: 01-12-2022 Through 02-12-2022, Dec. 2022. [Online]. Available: `https://www.scd.stfc.ac.uk/Pages/CIUK2022.aspx`.

[2] G. Mason-Williams, D. Bond, and M. Basham, *Distrac: Accelerating high performance compute processing for temporary data storage*, 2022. DOI: `10.48550/ARXIV.2212.03054`. [Online]. Available: `https://arxiv.org/abs/2212.03054`.

[3] ——, *DisTRaC - Distributed Transient Ram Ceph*, version 1.0.0, Sep. 2020. DOI: `10.5281/zenodo.4013776`. [Online]. Available: `https://doi.org/10.5281/zenodo.4013776`.

[4] G. Mason-Williams, *An Introduction to DosNA: Distributed Numpy Arrays for High-performance cloud computing*, version 1, Mar. 2022. DOI: `10.5281/zenodo.6411813`. [Online]. Available: `https://doi.org/10.5281/zenodo.6411813`.

[5] G. Mason-Williams, D. Bond, and M. Basham, *GRAM - General RAM*, version 1.0.0, Sep. 2020. DOI: `10.5281/zenodo.4014691`. [Online]. Available: `https://doi.org/10.5281/zenodo.4014691`.